Chapter 1 • Real Numbers

Solved Exercise 1.3

1. The sum of three consecutive integers is 42. Find them.

Let the integers be n, n+1, n+2.
3n + 3 = 42
⇒ 3n = 39
⇒ n = 13.
13, 14 and 15.

2. In right triangle ABC, AC = √3 + √5 cm and area = 1 + √15 cm². Find AB in the form a√3 + b√5.

AB = $\frac{2(1 + \sqrt{15})}{\sqrt{3} + \sqrt{5}}$.
AB = −(1 + √15)(√3 − √5)
= −√3 + √5 − 3√5 + 5√3.
AB = (4√3 − 2√5) cm.

3. A rectangle has sides 2 + √18 m and 5 − $\frac{4}{\sqrt{2}}$ m. Find its area in the form a + b√2.

2 + √18 = 2 + 3√2 and 5 − $\frac{4}{\sqrt{2}}$ = 5 − 2√2.
Area = (2 + 3√2)(5 − 2√2)
= 10 + 11√2 − 12.
(11√2 − 2) m².

4. Find two numbers whose sum is 68 and difference is 22.

x + y = 68 and x − y = 22. Adding gives 2x = 90, so x = 45.
y = 68 − 45
= 23.
45 and 23.

5. Convert 48°C to Fahrenheit using °F = ($\frac{9}{5}$)°C + 32.

°F = ($\frac{9}{5}$)(48) + 32
= 86.4 + 32.
118.4°F.

6. The sum of a father's and son's ages is 72. Six years ago, the father was twice the son's age. Find the son's age six years ago.

Let present ages be f and s. Then f + s = 72 and f − 6 = 2(s − 6).
Thus f = 2s − 6; substitution gives 3s − 6 = 72, so s = 26.
Six years ago: s − 6 = 20.
20 years.

7. Mirha bought a toy for Rs. 1500 and sold it for Rs. 1520. Find the profit percentage.

Profit = 1520 − 1500
= Rs. 20.
Profit % = ($\frac{20}{1500}$) × 100
= $\frac{4}{3}$%.
1.33% approximately.

8. Annual income is Rs. 960,000 and the exempt amount is Rs. 130,000. Find tax at 0.75%.

Taxable income = 960,000 − 130,000
= Rs. 830,000.
Tax = 0.0075 × 830,000.
Rs. 6,225.

9. Find compound markup on Rs. 375,000 for one year at 14% compounded annually.

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Markup = 375,000 × 0.14 × 1.
Rs. 52,500.

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